

Functions

Programmers love shortcuts that make writing code easier. One of the most common shortcuts is to give a name to a block of code that does an especially useful job. Then, instead of having to type out the whole block each time you need it, you simply type its name. These named blocks of code are called functions.

How to use a function

Using a function is also known as “calling” it. To call a function, you just type the function’s name, followed by a set of brackets that contain any parameters you want the function to work with. Parameters are a bit like variables that belong to the function, and they allow you to pass data between different parts of your program. When a function doesn’t need any parameters, the brackets are left empty.



LINGO

Function terms

There are a number of special words that coders use when talking about functions.

Call To use a function.

Define When you use the `def` keyword and write the code for a function, coders say you “define” the function. You also define a variable when you first set its value.

Parameter A piece of data (information) that you give to a function to use.

Return value Data that you pass from a function back to the main code. You get it using the keyword `return`.

Built-in functions

Python has a number of built-in functions that you can use in your code. These are helpful tools that let you do lots of tasks, from inputting information and showing messages on the screen to converting one type of data into another. You’ve already used some of Python’s built-in functions, such as `print()` and `input()`. Have a look at these examples. Why not try them out in the shell?

This asks the user to type in their name.

```
>>> name = input('What is your name?')
What is your name? Sara
>>> greeting = 'Hello' + name
>>> print(greeting)
Hello Sara
```

This shows the content of the variable `greeting` on the screen.

△ `input()` and `print()`

These two functions are like opposites. The `input()` function lets the user give instructions or data to the program by typing them in. The `print()` function sends output to the user by displaying messages or results on the screen.

